Feasibility of Locality-Sensitive Hashing Filters for Data Leakage Prevention*

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Abstract

We evaluate the Locality-Sensitive Hashing Filter (LSHF) as an offline pre-filtering mechanism for Data Leakage Prevention. By utilizing Content-Defined Chunking and dual-threshold tuning, LSHF effectively identifies suspicious similarities locally with near-zero false negatives. Our analysis shows it filters up to 90% of benign queries, providing a highly efficient solution for resource-constrained security environments.

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